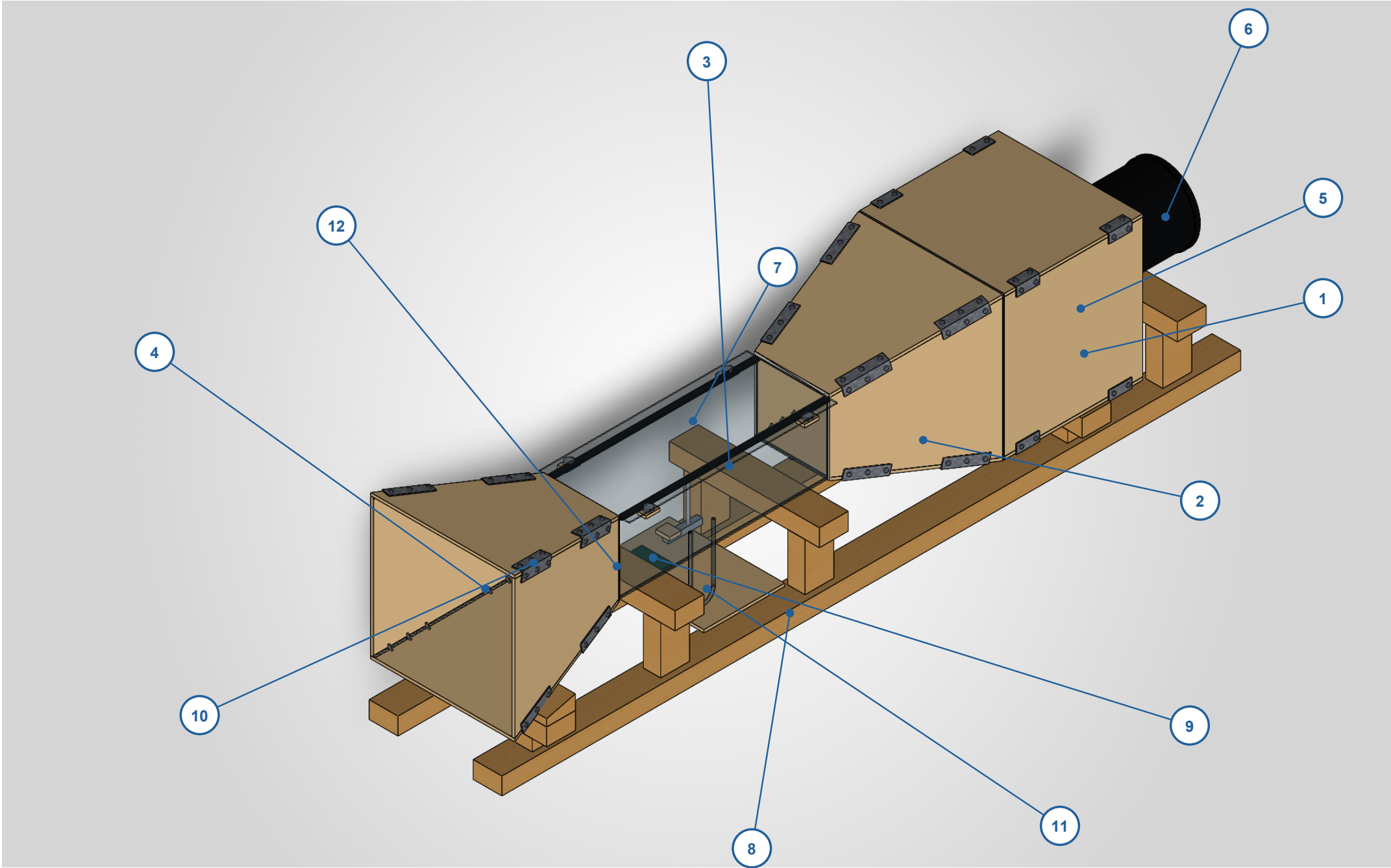


Labeled assembly

The final SolidWorks model, including the diffuser corner braces and separate caulk components.



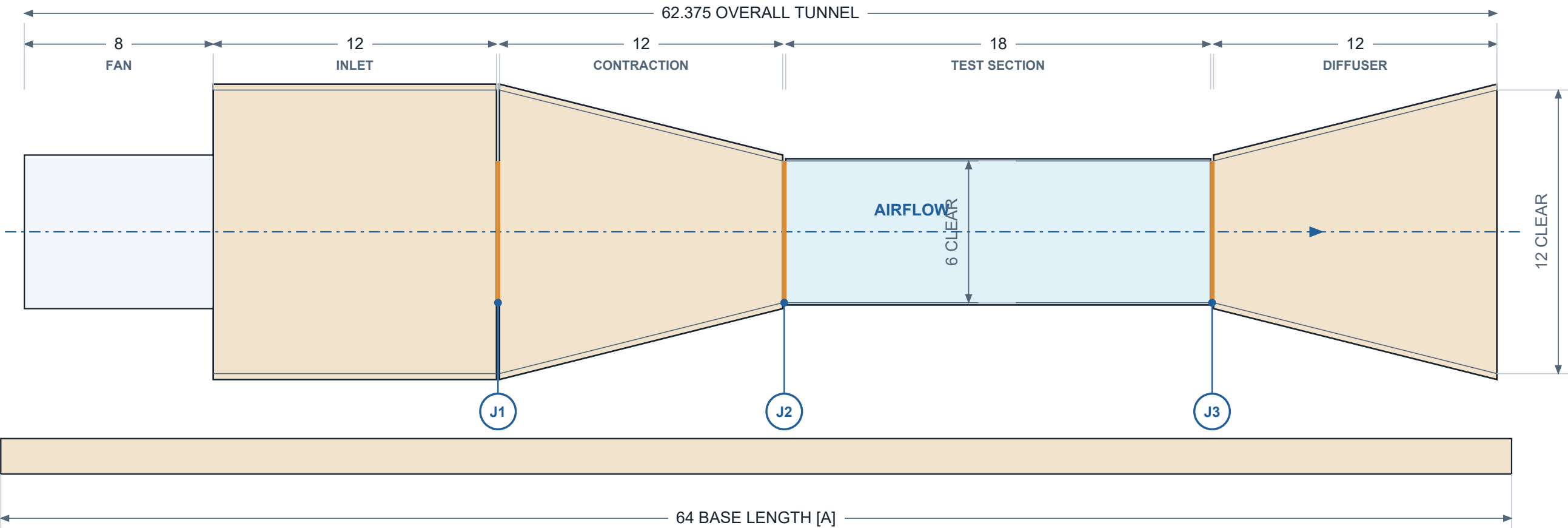
- 01 Inlet panels**
P01-P04 | 12 x 12 in clear
- 02 Contraction panels**
P05-P08 | 12 -> 6 in
- 03 Acrylic chamber + lid**
P09-P12 | 18 in long
- 04 Diffuser panels**
P13-P16 | 6 -> 12 in
- 05 Flow conditioning (inside)**
P17-P19, P40 | honeycomb + screens
- 06 Inline fan**
P20-P22 | simplified housing / rotor
- 07 Lid mounting hardware**
P23-P25 | gasket, tabs, magnets
- 08 Base and supports**
P26-P31 | proposed timber layout
- 09 Measurement assembly**
P32-P36 | rod, load cell, electronics
- 10 Inside corner braces**
P37-P38 | 16 large + 8 small
- 11 Additional hardware**
P41-P43 | screws, U-tube, saddle
- 12 Applied caulk components**
P44-P50 | 17 separate instances

49 installed part types / 234 instances / 1 additional measured reference panel (P39)

Section dimensions & layout

Dimensioned schematic of the CAD reconstruction. Views are not to scale; all dimensions are inches unless marked mm.

SIDE ELEVATION / AIRFLOW +X



SECTION JOINTS [A]

J1, J2 and J3 each contain a 0.125 axial caulk gap. The test module is also offset +0.125 transversely. Exact as-built gaps were not measured.

TRANSVERSE CLEARANCES

Inlet and diffuser exit: 12 x 12. Test passage: 6 x 6 nominal. Plywood radial wall: 0.250; acrylic: 0.093. Lid: 7.8 wide [A].

DOCUMENTED VS ASSUMED

Unmarked main section sizes come from project drawings / BOM. [A] marks additional layout assumptions. Base envelope: 64 x 13 x 16.66 [A].

Panels & removable lid

Projected panel profiles of the native CAD parts. Tapered profiles are not manufacturing flat patterns.

INLET / TOP + BOTTOM

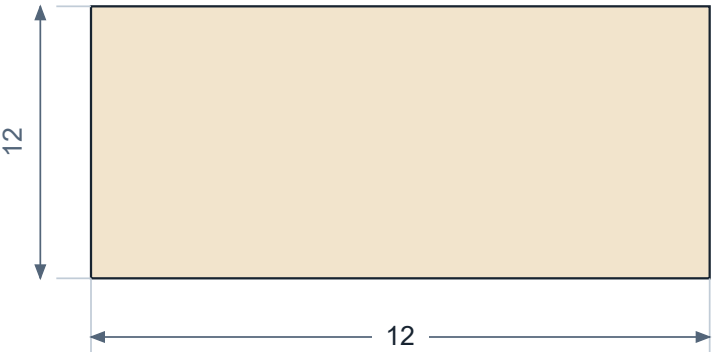
P01-P02 | Qty 2 | BCX plywood 0.250



Top / bottom lap over side-panel edges.

INLET / SIDES

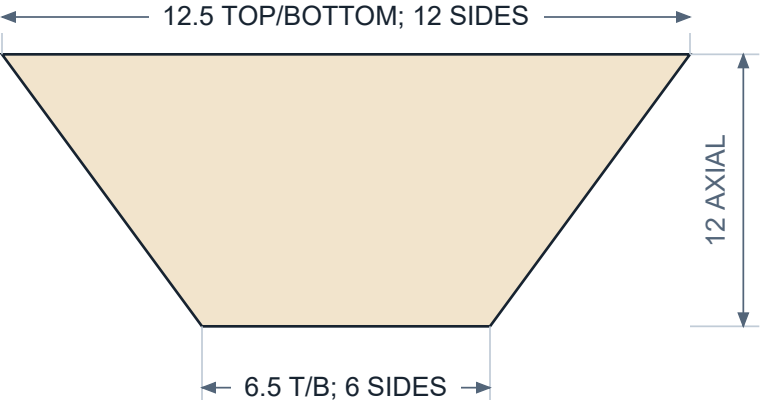
P03-P04 | Qty 2 | BCX plywood 0.250



12 in nominal clear inlet.

TAPERED WOOD PANELS

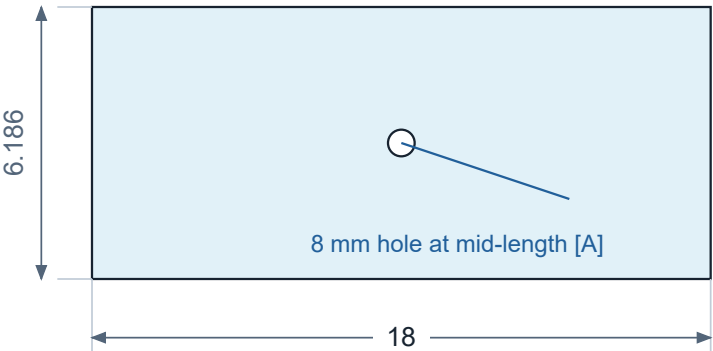
P05-P08 + P13-P16 | Qty 8



Reverse the profile for the diffuser.

ACRYLIC FLOOR

P09 | Qty 1 | Acrylic 0.093



Rod clearance is a model assumption.

ACRYLIC SIDES

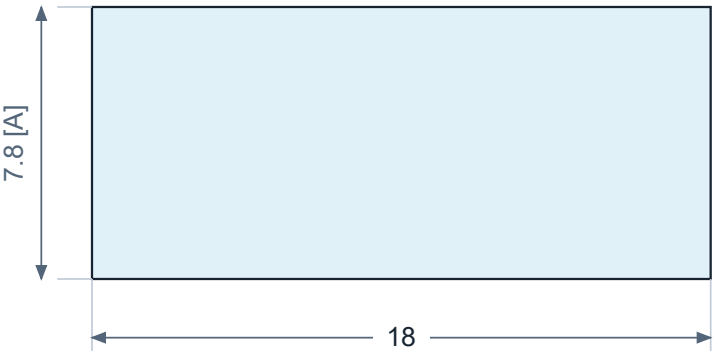
P10-P11 | Qty 2 | Acrylic 0.093



0.125 gasket fills remaining side height [A].

REMOVABLE LID

P12 | Qty 1 | Acrylic 0.093

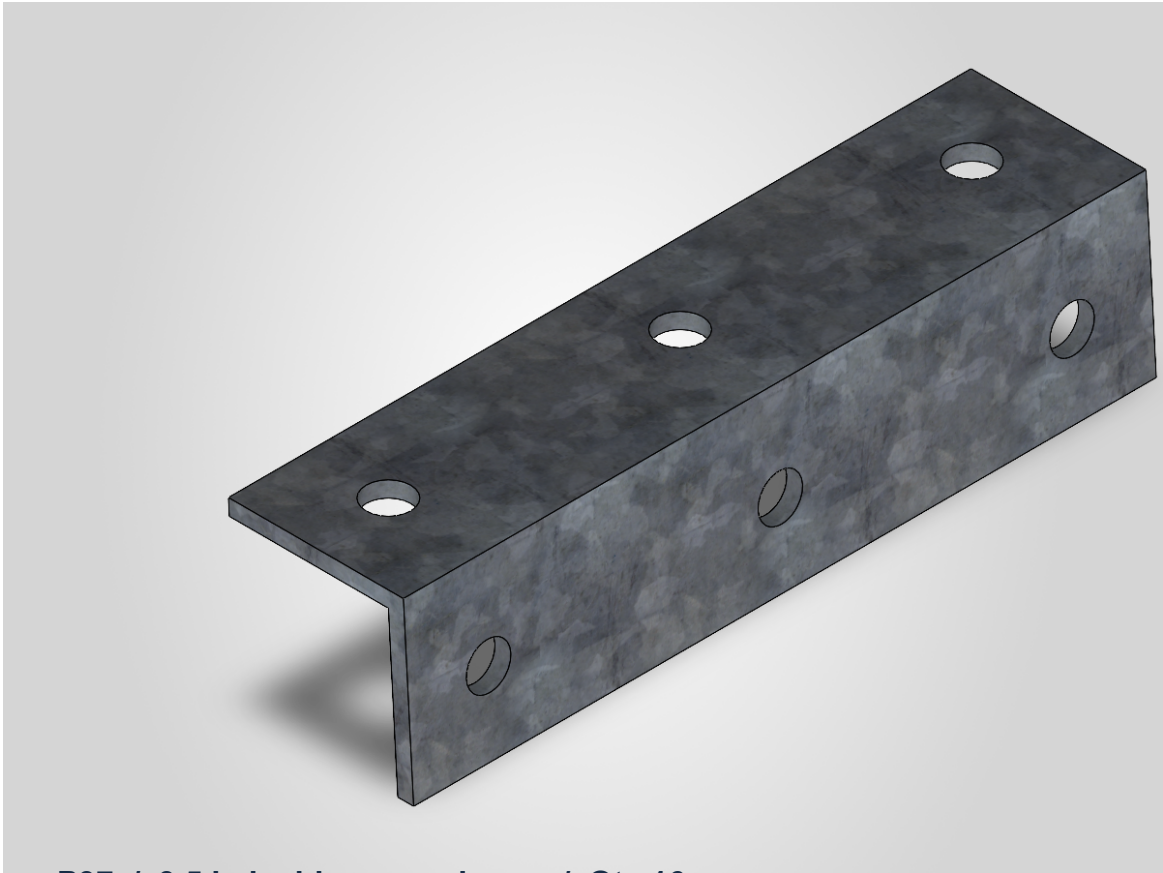


Width covers the modeled magnetic tabs [A].

Measured asymmetric panel P39 is supplied separately: 12.00 / 5.75 end widths; 12.5 / 12.375 edges. It is not installed in the nominal assembly.

Inside corner braces

Longitudinal folded corner braces from the hardware photos, repeated on the diffuser at the user's request.



P37 / 3.5 in inside corner brace / Qty 16

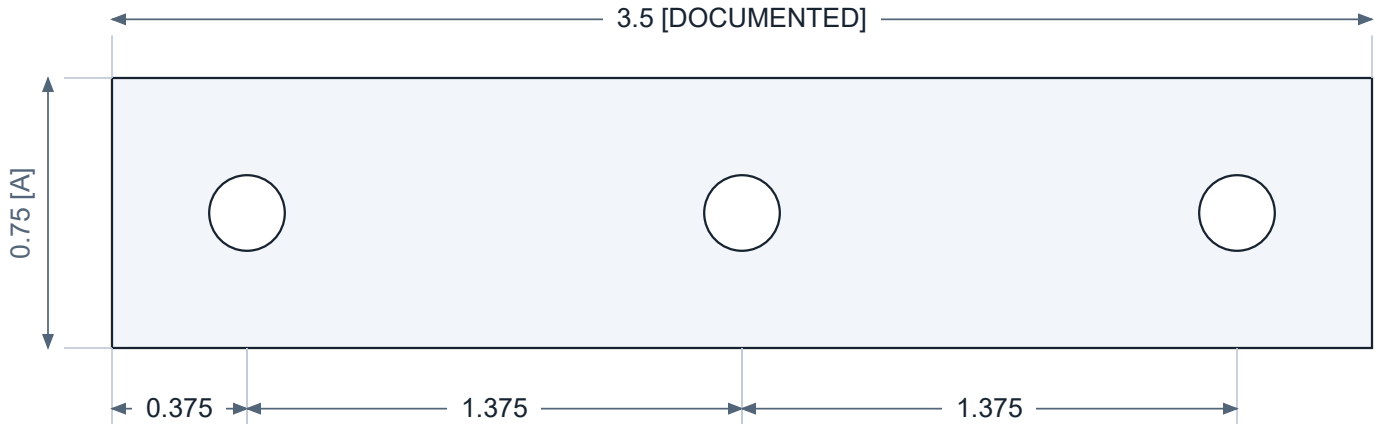
Two per corner on contraction and diffuser.

BEND / MATERIAL

P37 has an assumed installed 93.37 degree bend to follow the tapered seam. P38 uses 90 degrees. Zinc-plated steel is represented by a galvanized-steel material proxy.

FLANGE ELEVATION

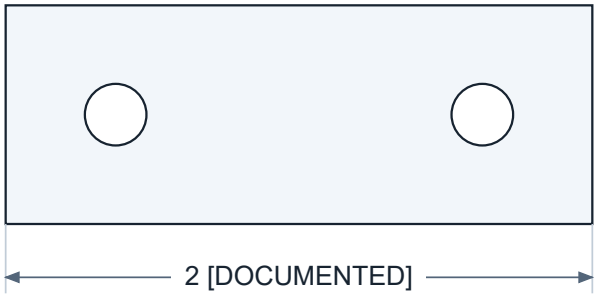
Holes and flange width are assumptions; stamped ribs / countersinks omitted.



6 x 0.21 dia. holes [A] total, 3 per flange. Gauge: 0.075 [A].

P38 / 2 in BRACE

Qty 8 on inlet; 4 holes total, 2 per flange [A].



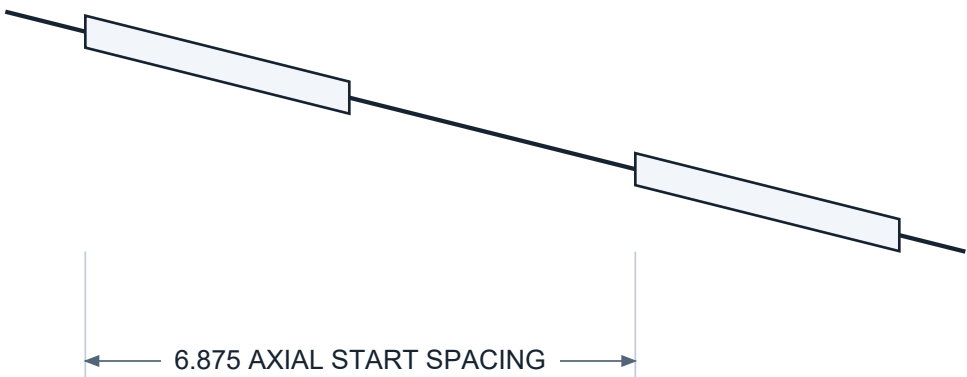
Flanges 0.75; gauge 0.075 [A].

Large brace: 8 contraction + 8 diffuser.

P41 screws: #8 x 3/4 in, 128 instances.

REPEATED SEAM SPACING

Axial locations from each tapered section start [A].



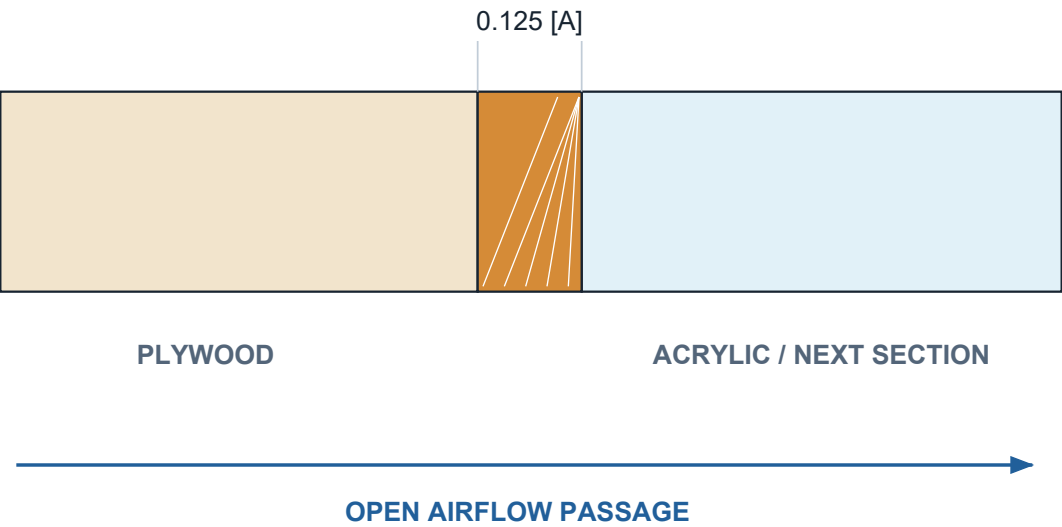
Starts: 1.000 and 7.875. Section length: 12.

Caulk joints & imperfect fit

Independent sealant parts represent the fit-up gaps. Their exact shape and size remain assumptions.

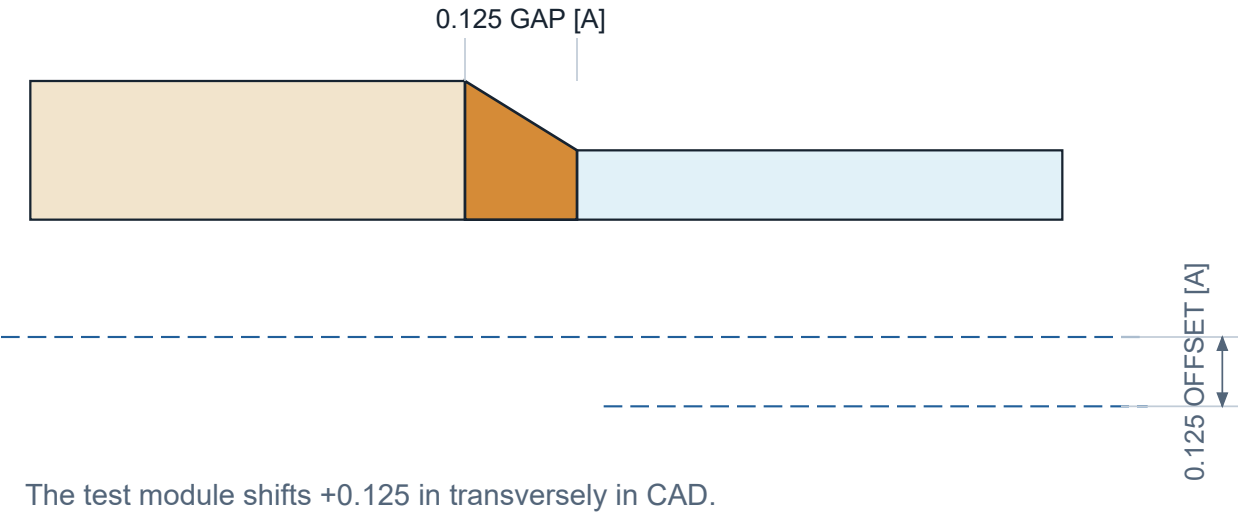
A / AXIAL SECTION GAP

P48-P50: three hollow joint rings; airflow passage remains open.



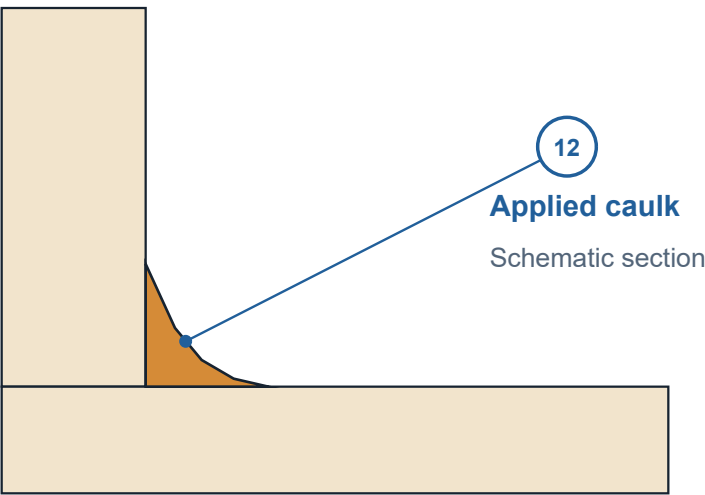
B / TRANSVERSE MISMATCH

Plan detail at acrylic joint, exaggerated for clarity.



C / INTERNAL CORNER BEAD

P44-P47: 1/8 in plywood seam beads; 1/16 in acrylic floor beads [A].



MODEL SCOPE / PART KEY

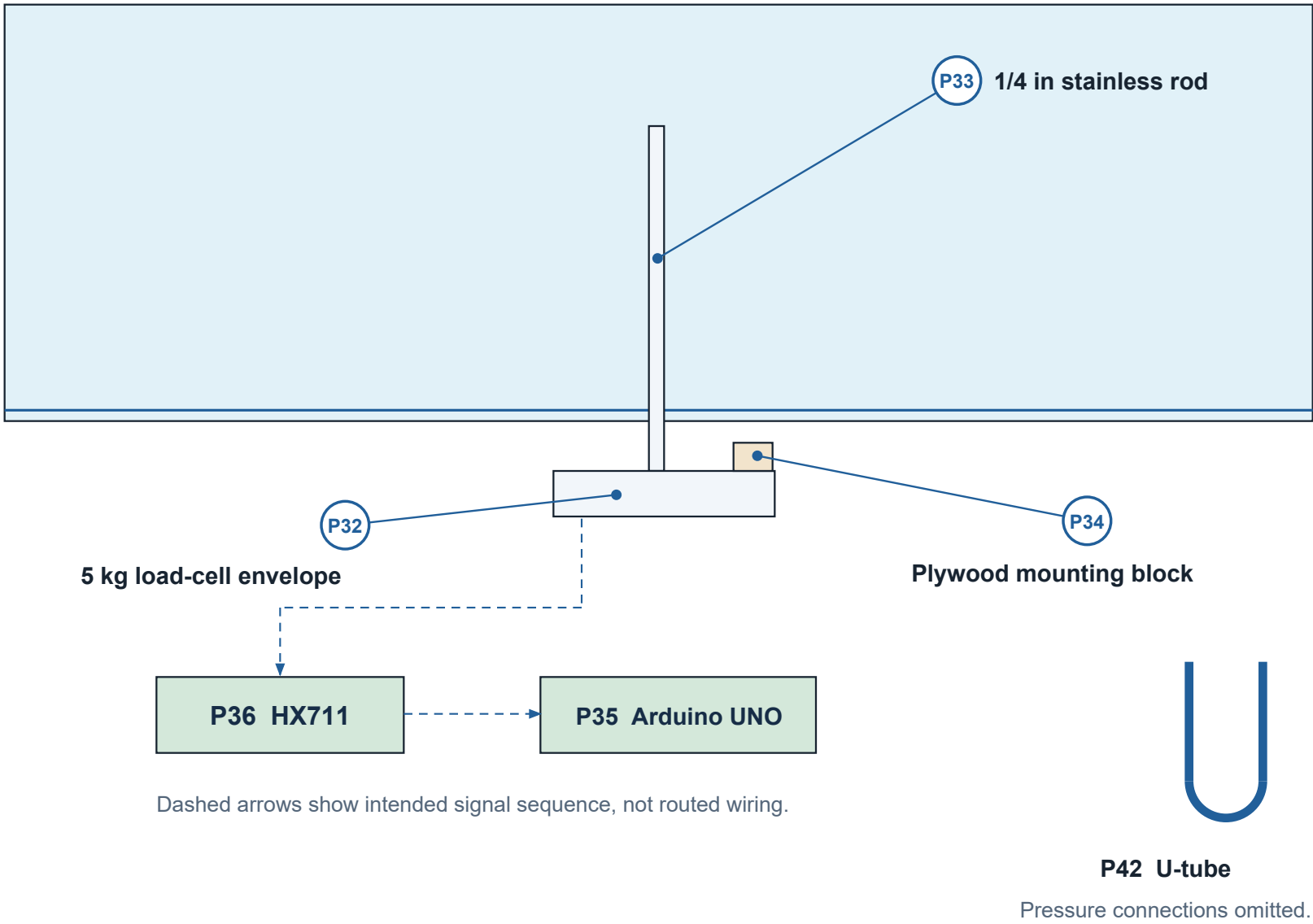
P44 inlet corner beads x4; P45 contraction x4; P46 diffuser x4; P47 acrylic floor seams x2. P48, P49 and P50 bridge the three module joints. White caulk is shown amber on this sheet for visibility. The removable lid uses gasket rails P23; its release and end seals need physical verification. Sealing, adhesion and airtightness are not proven by the CAD reconstruction.

Instrumentation & support parts

Component locations and simplified envelopes, with a schematic signal path for the measurement system.

TEST-SECTION DETAIL

Diagrammatic side section; electronics and load-cell envelopes are simplified.



TIMBER / SUPPORT KEY

Assumed frame dimensions, inches.

P26	Base rails x2
	64 x 2.5 x 1.5
P27	Crossbeams x5
	13 x 2.5 x 1.5
P28	Short posts x4
	1.5 x 2.5 x 1.0
P29	Test posts x4
	1.5 x 2.5 x 4.157
P30	Fan posts x2
	1.5 x 2.5 x 4.125
P31	Electronics shelf
	10 x 7 x 0.25
P43	Diffuser saddle
	13 wide; 2.5 axial

STOCK CHECK

The proposed support layout uses about 19.6 ft before kerf, exceeding the 16 ft purchased. Additional stock or a revised frame is required.

Complete parts key / 1 of 2

P-numbers are drawing callouts. Filename stems below map directly to the individual .SLDPRT files.

ID	NATIVE PART FILENAME (without .SLDPRT)	QTY	MATERIAL	MODEL ROLE / NOTE
P01	01_Inlet_Top	1	BCX plywood	Separate native component / repeated as listed.
P02	01_Inlet_Bottom	1	BCX plywood	Separate native component / repeated as listed.
P03	01_Inlet_Right	1	BCX plywood	Separate native component / repeated as listed.
P04	01_Inlet_Left	1	BCX plywood	Separate native component / repeated as listed.
P05	02_Contraction_Top	1	BCX plywood	Separate native component / repeated as listed.
P06	02_Contraction_Bottom	1	BCX plywood	Separate native component / repeated as listed.
P07	02_Contraction_Right	1	BCX plywood	Separate native component / repeated as listed.
P08	02_Contraction_Left	1	BCX plywood	Separate native component / repeated as listed.
P09	03_Test_Floor	1	Acrylic	Separate native component / repeated as listed.
P10	03_Test_Right	1	Acrylic	Separate native component / repeated as listed.
P11	03_Test_Left	1	Acrylic	Separate native component / repeated as listed.
P12	03_Removable_Lid	1	Acrylic	Lid 7.8 x 18 x 0.093 in [A].
P13	04_Diffuser_Top	1	BCX plywood	Separate native component / repeated as listed.
P14	04_Diffuser_Bottom	1	BCX plywood	Separate native component / repeated as listed.
P15	04_Diffuser_Right	1	BCX plywood	Separate native component / repeated as listed.
P16	04_Diffuser_Left	1	BCX plywood	Separate native component / repeated as listed.
P17	05_Aramid_Honeycomb	1	Aramid	300 x 300 x 15 mm; pores representative.
P18	05_Aluminum_Screen	2	1060 Alloy	Separate native component / repeated as listed.
P19	05_Honeycomb_Edge_Seal	1	Foam strip	Separate native component / repeated as listed.
P20	06_Fan_Housing	1	Steel	Separate native component / repeated as listed.
P21	06_Fan_End_Collar	2	Steel	Separate native component / repeated as listed.
P22	06_Fan_Impeller	1	ABS	Separate native component / repeated as listed.
P23	07_Lid_Gasket_Rail	2	Foam strip	Separate native component / repeated as listed.
P24	07_Magnet_Support_Tab	4	BCX plywood	Separate native component / repeated as listed.
P25	07_Disc_Magnet	8	Nd magnet	Separate native component / repeated as listed.
P26	08_Base_Rail	2	Pine	64 in assumed length; two rails.

TOTAL: 50 part files | 49 installed part types | 234 assembly instances | 1 reference panel

Complete parts key / 2 of 2

P-numbers are drawing callouts. Filename stems below map directly to the individual .SLDPRT files.

ID	NATIVE PART FILENAME (without .SLDPRT)	QTY	MATERIAL	MODEL ROLE / NOTE
P27	08_Crossbeam	5	Pine	Separate native component / repeated as listed.
P28	08_Inlet_Diffuser_Post	4	Pine	Separate native component / repeated as listed.
P29	08_Test_Support_Post	4	Pine	Separate native component / repeated as listed.
P30	08_Fan_Support_Post	2	Pine	Separate native component / repeated as listed.
P31	08_Electronics_Shelf	1	BCX plywood	Separate native component / repeated as listed.
P32	09_Load_Cell_5kg	1	1060 Alloy	5 kg rating; simplified envelope.
P33	09_Stainless_Model_Rod	1	AISI 304	Separate native component / repeated as listed.
P34	09_Load_Cell_Mount	1	BCX plywood	Separate native component / repeated as listed.
P35	09_Arduino_UNO_Board	1	PCB laminate	Separate native component / repeated as listed.
P36	09_HX711_Board	1	PCB laminate	Separate native component / repeated as listed.
P37	10_Corner_Brace_3_5in	16	Zinc-steel proxy	8 contraction + 8 diffuser; 3.5 in long.
P38	10_Corner_Brace_2in	8	Zinc-steel proxy	8 inlet; 2 in long. Hidden repeats assumed.
P39	REF_Measured_Trapezoid_A	0	BCX plywood	Measured reference only; not installed.
P40	11_Screen_Retaining_Frame	2	BCX plywood	Separate native component / repeated as listed.
P41	11_Wood_Screw_No8_3_4in	128	Zinc-steel proxy	128 modeled screws; not purchase count.
P42	11_Manometer_U_Tube	1	Clear vinyl	Separate native component / repeated as listed.
P43	11_Diffuser_Saddle	1	Pine	Separate native component / repeated as listed.
P44	12_Caulk_Inlet_Corner	4	Caulk	Separate native component / repeated as listed.
P45	12_Caulk_Contraction_Corner	4	Caulk	Separate native component / repeated as listed.
P46	12_Caulk_Diffuser_Corner	4	Caulk	Separate native component / repeated as listed.
P47	12_Caulk_Acrylic_Floor_Seam	2	Caulk	Separate native component / repeated as listed.
P48	12_Caulk_Inlet_Contraction_Joint	1	Caulk	Separate native component / repeated as listed.
P49	12_Caulk_Contraction_Acrylic_Joint	1	Caulk	Two solid bodies form this gap bridge.
P50	12_Caulk_Acrylic_Diffuser_Joint	1	Caulk	Separate native component / repeated as listed.

TOTAL: 50 part files | 49 installed part types | 234 assembly instances | 1 reference panel

Basis: project BOM and drawings, measured panel photo 10, corner-brace package photo 03, assembly photos 13/21, and the completed native model. CAD quantities include repeated hardware assumptions. See Parts_register.csv and README.md for the full dimensional basis.